Year 2 Statistics and Mechanics

Chapter 5

Forces and friction

Practice Booklet

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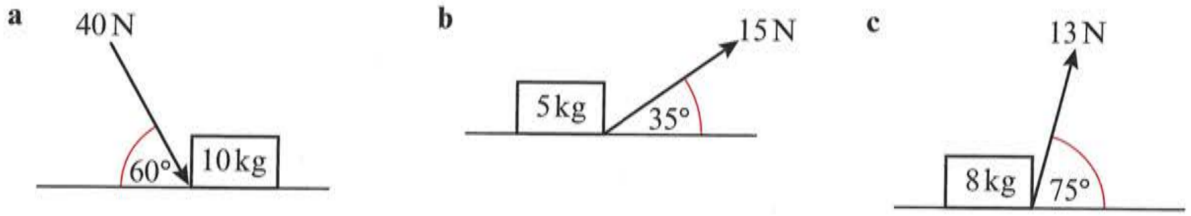
5.1 Resolving forces

Practice Book, printed page 43.

1. Each diagram shows a box on a smooth surface acted on by a single force.
In each case, work out:

i. the acceleration of the box

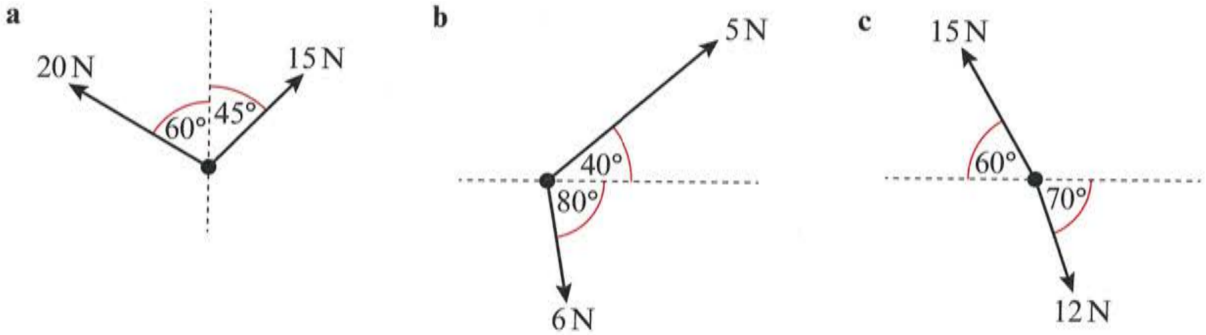
ii. the normal reaction between the box and the floor.



Hint

To find the normal reaction between the box and the floor, resolve vertically, considering the weight of the box and the vertical component of the force acting on the box.

2. Each diagram shows a particle being acted upon by two forces. In each case, work out the magnitude and direction of the resultant force acting on the particle.



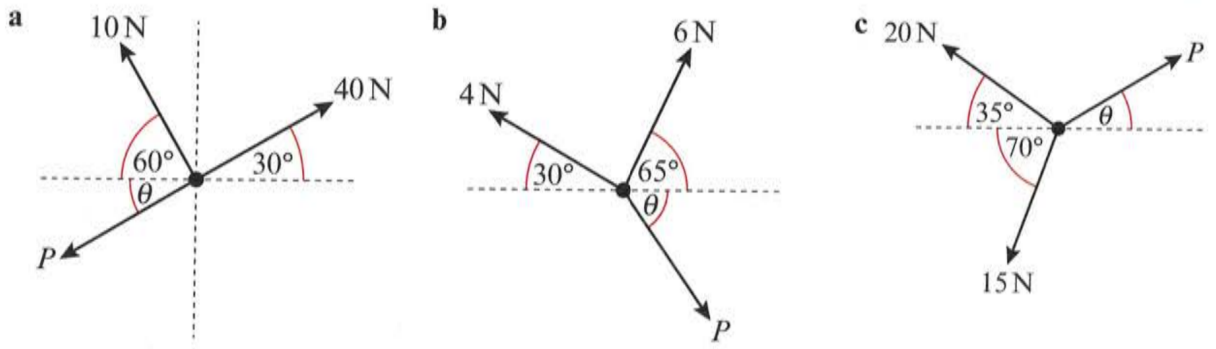
Hint

You can use the triangle law for vector addition. The resultant force will be the third side of the triangle formed by adding the two given forces end-to-end.

3. Each diagram shows a particle in equilibrium, acted upon by three forces.
In each case, work out:

i. the size of the angle θ

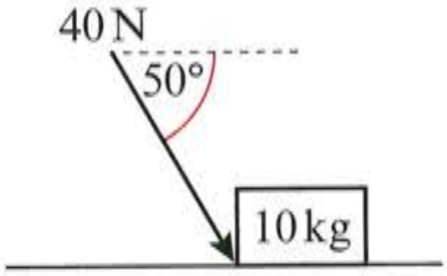
ii. the magnitude of P.



Hint

The particles are in equilibrium, so the resultant force is zero. Resolve horizontally and vertically, then solve the equations simultaneously to find P and θ.

4. A box of mass 10 kg lies on a smooth horizontal floor. A force of 40 N is applied to the box at an angle of 50° below the horizontal, as shown in the diagram.
Work out:



a. the acceleration of the box

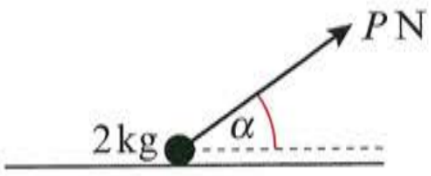
b. the normal reaction between the box and the floor.

5. E A box of mass m kg sits at rest on a smooth horizontal surface. A force of 20 N is applied to the box at an angle of 30° above the horizontal, causing the box to accelerate at 3 m s^{-2}.

a. Work out the value of m. (2 marks)

b. State what effect the assumption that the surface is smooth has on your calculations. (1 mark)

6. E/P A particle of mass 2 kg sits at rest on a smooth horizontal surface. A force of P N is applied to the particle at an angle of α above the horizontal, where $\tan\alpha = \frac{3}{4}$



Given that the particle travels 40 m in the first 5 seconds of its motion, work out the value of P. (4 marks)

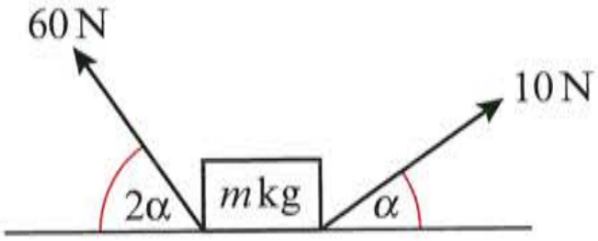
7. E/P A box of mass 3 kg sits at rest on a smooth horizontal surface. A force of 6 N is applied to the box at an angle θ above the horizontal, reducing the normal reaction between the box and the surface by 3 N.

a. Work out the size of angle θ. (2 marks)

An additional force P, acting at an angle of 2θ above the horizontal, is applied to the box causing the acceleration of the box to double.

b. Show that $P = 6\sqrt{3}$ N. (4 marks)

8. E/P A box of mass m kg lies at rest on a smooth horizontal surface. The box is acted upon by two forces of magnitudes 10 N and 60 N, acting in opposite directions at angles of α and 2α above the horizontal respectively.



Find:

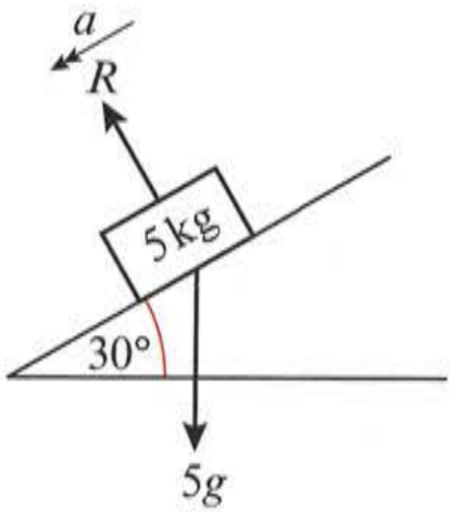
a. the size of α (7 marks)

b. the least possible value of m. (3 marks)

5.2 Inclined planes

Practice Book, printed page 45.

1. A box of mass 5 kg slides down a smooth plane that is inclined at 30° to the horizontal.
Work out:



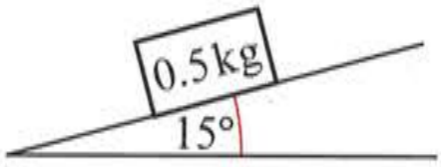
a. the normal reaction between the box and the plane

b. the acceleration of the box.

Hint

The normal reaction will be equal to the component of the weight of the box that acts perpendicular to the plane. The acceleration will be due to the component of the weight of the box that acts parallel to the plane.

2. A box of mass 0.5 kg lies on a smooth plane that is inclined at 15° to the horizontal. A force of magnitude 2 N is applied to the box parallel to and up the plane.



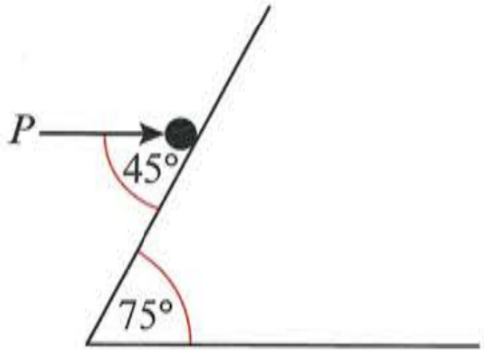
a. Copy and complete the diagram to show all the forces acting on the box.

b. Work out the acceleration of the box.

Hint

There are three forces acting on the box.

3. A particle of mass 0.25 kg lies on a smooth plane that is inclined at 75° to the horizontal. It is held at rest by a force P, applied to the particle at an angle of 45° to the plane as shown in the diagram.



a. Work out the magnitude of P.

The force P is removed.

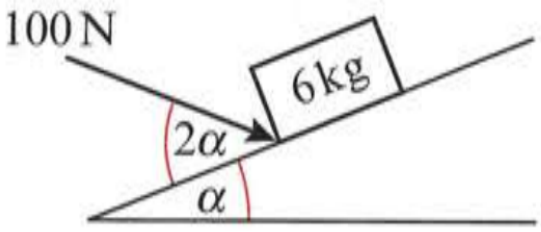
b. Work out the velocity of the particle after 2 seconds.

Hint

For part **b**, work out the acceleration of the particle then use $v = u + at$
← Year 1, Section 9.3

4. E/P A particle of mass m kg lies on a smooth plane that is inclined at an angle θ to the horizontal. Show that the acceleration of the particle is independent of its mass. (2 marks)

5. E/P A box of mass 6 kg sits on a smooth plane that is inclined at an angle α to the horizontal where $\tan\alpha = \frac{5}{12}$. A force of 100 N is applied to the box at an angle of 2α to the plane, causing the box to accelerate from rest up the hill.



After 2 seconds the force is removed. Calculate:

a. the speed of the box at the instant the force is removed (4 marks)

b. the length of time it takes the box to come to rest from the instant the force is removed. (4 marks)

6. E/P A box of mass m kg lies on a smooth plane that is inclined at 30° to the horizontal. A force of g N is applied to the box parallel to and up the plane. The box accelerates down the plane at $\frac{1}{3}g$ m s^{-2}. Work out the value of m. (4 marks)

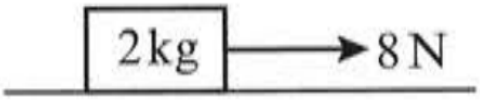
7. E/P A particle of mass 5 kg sits on a smooth plane that is inclined at 30° to the horizontal. A force of $2g$ N is applied vertically upwards to the particle. Calculate the magnitude of the acceleration of the particle and state its direction. (6 marks)

8. E/P A particle of mass 2 kg sits on a smooth plane that is inclined at 20° to the horizontal. The particle is initially at rest at point P, when a force of g N is applied to the particle parallel to and up the plane, causing it to accelerate up the plane. After 3 seconds the force is removed. Calculate how long, from the instant it first moves, it takes the box to return to P. (12 marks)

5.3 Friction

Practice Book, printed page 46.

1. The diagram shows a box lying on a rough horizontal plane. The box is acted upon by a horizontal force of magnitude 8 N, which causes it to accelerate.



a. Work out the magnitude of the frictional force and the acceleration of the box in the case when the coefficient of friction between the box and the plane is equal to:

i. 0.2

ii. 0.35

Hint

$F_{MAX} = \mu R$, where R is the normal reaction between the box and the plane, and μ is the coefficient of friction.

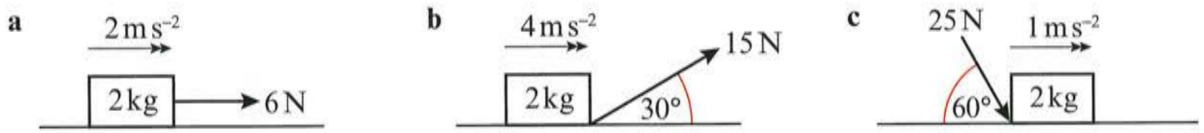
A student claims that in the case when $\mu = 1$, the magnitude of the frictional force will be $2g$ N and the box will remain at rest.

b. Explain why the student is incorrect, and state the magnitude of the frictional force in this case.

Hint

If the box is in equilibrium, the frictional force will only be as large as is necessary to prevent the box from moving.

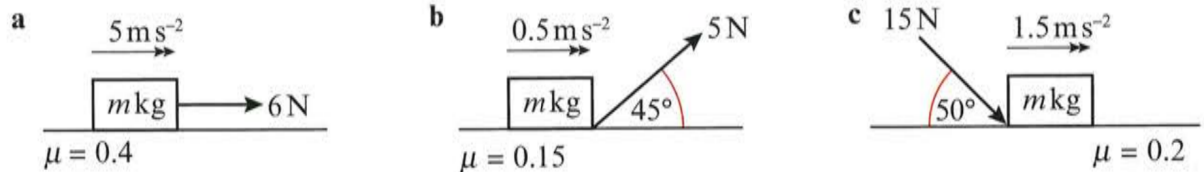
2. Each of the diagrams shows a box accelerating on a rough horizontal plane. In each case, work out the value of the coefficient of friction, μ, between the box and the plane.



Hint

In parts **b** and **c** you will need to consider the vertical component of the force acting on the box when working out the normal reaction.

3. Each of the diagrams shows a box accelerating on a rough horizontal plane.
The coefficient of friction between the box and the plane is μ.
In each case, work out the value of m.



4. A box of mass 2 kg lies on a rough plane that is inclined at 30° to the horizontal. The coefficient of friction between the box and the plane is 0.1. A force of 15 N is applied to the box parallel to and up the plane. Work out the acceleration of the box.

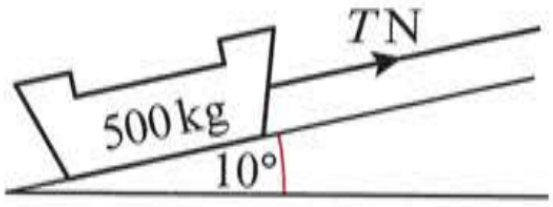
5. E A block of mass 0.2 kg sits at rest on a rough horizontal surface. The coefficient of friction between the surface and the block is 0.4. A horizontal force of 4 N is applied to the block. After 5 seconds the force is removed. Calculate:

a. the speed of the box at the instant the force is removed (3 marks)

b. the length of time it takes the box to come to rest from the instant the force is removed. (3 marks)

6. E/P A particle of mass m kg is sliding down a rough slope inclined at an angle of θ to the horizontal. The coefficient of friction between the particle and the slope is μ.
Show that the acceleration of the particle is given by $a = g(\sin\theta - \mu\cos\theta)$ (4 marks)

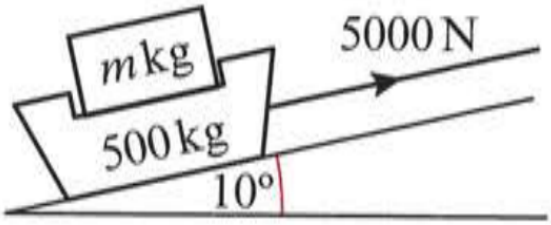
7. E/P A sled of mass 500 kg sits at rest on a rough slope that is inclined at 10° to the horizontal. The coefficient of friction between the slope and the sled is 0.3.
The sled is held in place by a wire attached to a winch. The tension in the wire is T N.



The sled is modelled as a particle, and the wire is modelled as a light inextensible string.

a. Find the maximum possible value of T. (3 marks)

A load of mass m kg is added to the sled, and the tension in the wire is increased to 5000 N in order to pull the sled and load up the slope.



Given that the sled and load are pulled a distance of 100 m in 80 seconds,

b. work out the value of m. (6 marks)

Problem solving: Set A

Bronze

A box of mass 3 kg sits at rest on a rough horizontal surface. The coefficient of friction between the surface and the box is 0.15. A horizontal force of 6 N is applied to the block.

a. Draw a diagram to show all the forces acting on the box. (2 marks)

b. Calculate the distance travelled by the block in the first 6 seconds of its motion. (5 marks)

Silver

A block of mass 2 kg sits at rest on a rough horizontal surface. The coefficient of friction between the surface and the block is 0.25. A force of 10 N that acts at 60° above the horizontal is applied to the block.
Calculate the velocity of the block 10 seconds after it begins to move. (5 marks)

Gold

A block of mass m kg sits at rest on a rough horizontal surface. The coefficient of friction between the surface and the block is μ. A force of $2g$ N pulls the block at an angle of θ above the horizontal.
Show that the velocity of the block after t seconds is given by $\left(\frac{2\cos\theta + 2\mu\sin\theta}{m} - \mu\right)gt$ (5 marks)

Problem solving: Set B

Bronze

A block of mass 0.2 kg lies on a rough slope that is inclined at 30° to the horizontal.
The coefficient of friction between the block and the slope is 0.2. A force of P N acts on the block in a direction parallel to and up the slope, causing it to accelerate up the slope.

a. Draw a diagram to show all the forces acting on the box. (2 marks)

b. Work out the exact magnitude of the frictional force. (1 mark)

c. Given that the block accelerates up the slope at 0.2 m s^{-2}, work out the value of P. (3 marks)

Silver

A particle of mass m kg sits at rest on a rough slope that is inclined at an angle α to the horizontal where $\tan\alpha = \frac{3}{4}$
The coefficient of friction between the particle and the slope is 0.1. A force of magnitude F N acts on the particle in a direction parallel to and up the slope.
Given that the particle accelerates up the slope,

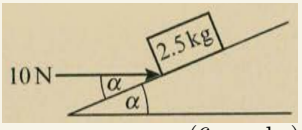
a. show that $F > kmg$, where k is a constant to be determined. (3 marks)

Given further that $F = 50$ and that the particle accelerates up the slope at 1.1 m s^{-2},

b. work out the value of m. (3 marks)

Gold

A box of mass 2.5 kg sits at rest on a rough slope that is inclined at an angle α to the horizontal. The coefficient of friction between the box and the slope is 0.15.
A force of magnitude 10 N acts horizontally on the box.

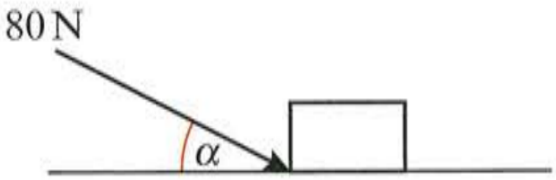


Given that the box is in limiting equilibrium, work out the two possible values of α. (6 marks)

Selected exam question bank questions

Question 22

A box is pushed along a floor. The floor is modelled as a rough horizontal plane and the box is modelled as a particle. The coefficient of friction between the box and the floor is 0.75.



The box is pushed by a force of magnitude 80 N that acts downwards at an angle of α with the floor, where $\tan\alpha = \frac{5}{12}$

Given that the box moves with constant speed, find the mass of the box. (5 marks)

Question 25

A box of mass 60 kg is moving along a horizontal floor under the action of a force of magnitude P N. The force acts at 60° above the floor and the box moves in a straight line at constant speed.

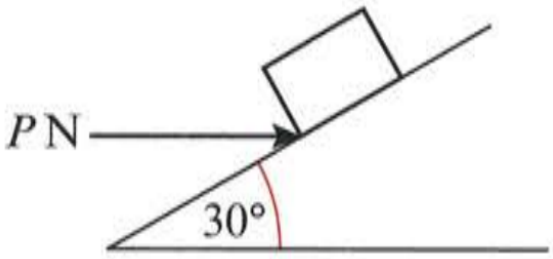
By modelling the box as a particle and the floor as a rough horizontal plane, with coefficient of friction 0.25 between the box and the plane,

a. calculate the value of P. (5 marks)

b. Without further calculation, briefly explain how the value of P would change if the surface is modelled as being smooth. (1 mark)

Question 29

A box of weight 20 N lies on a rough plane inclined at an angle of 30° to the horizontal. A horizontal force of magnitude P N acts on the box.



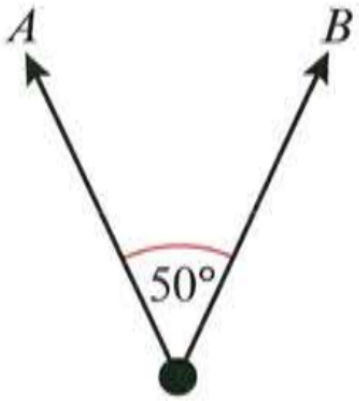
The box is in equilibrium and on the point of slipping up the plane. The normal reaction of the plane on the box is 32 N, and the coefficient of friction between the box and the plane is μ. Find:

a. the value of P (3 marks)

b. the value of μ. (3 marks)

Question 32

Two forces, A and B, act on a particle. Force A has magnitude 6 N and force B has magnitude 5 N. The angle between the directions of A and B is 50°.



Work out:

a. the magnitude of the resultant of the two forces (3 marks)

b. the angle between force A and the resultant. (2 marks)

Question 36

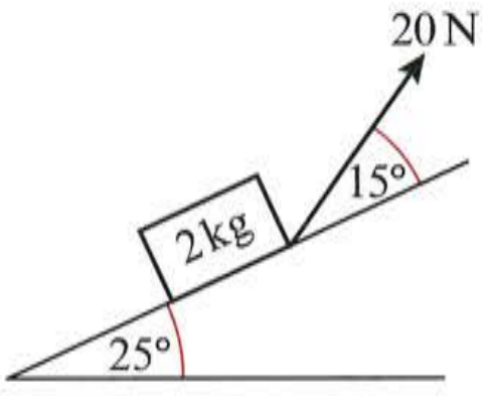
A sledge of mass 30 kg is pulled up a rough plane by means of a rope that is parallel to the line of greatest slope of the plane. The plane is inclined at an angle α to the horizontal, where $\tan\alpha = \frac{3}{4}$

The coefficient of friction between the sledge and the plane is 0.2, and the sledge is accelerating up the slope with acceleration 1.25 m s^{-2}.

Find the tension in the rope. (6 marks)

Question 47

A block P of mass 2 kg is moving up a rough plane at a constant speed of 8 m s^{-1} under the action of a force of magnitude 20 N. The plane is inclined at 25° to the horizontal, and the force acts at 15° to the inclined plane as shown in the diagram.



The coefficient of friction between the block and the plane is μ.

a. Work out the value of μ. (6 marks)

The force of magnitude 20 N is removed.

b. Find the distance that P travels from the instant when the force was removed to the instant when it comes to rest. (4 marks)

Answers

5.1 Resolving forces

1.

a.

i. 2 m s^{-2}

ii. 133 N (3 s.f.)

b.

i. 2.46 m s^{-2}

ii. 40.4 N (3 s.f.)

c.

i. 0.421 m s^{-2}

ii. 65.8 N (3 s.f.)

2.

a. 21.7 N (3 s.f.) at an angle of 18.0° (3 s.f.) to the left of the +**j** vector

b. 5.57 N (3 s.f.) at an angle of 28.9° (3 s.f.) below the +**i** vector

c. 3.80 N (3 s.f.) at an angle of 26.8° (3 s.f.) above the −**i** vector

3.

a.

i. 44.0° (3 s.f.)

ii. 41.2 N (3 s.f.)

b.

i. 82.9° (3 s.f.)

ii. 7.50 N (3 s.f.)

c.

i. 6.95° (3 s.f.)

ii. 21.7 N (3 s.f.)

4.

a. 2.57 m s^{-2}

b. 129 N

5.

a. $\frac{10\sqrt{3}}{3}$

b. No resistance due to surface friction.

6. 8

7.

a. 30°

b. For the initial acceleration of the box:

$$\mathrm{R}(\rightarrow): \ 6\cos 30^\circ = 3a \text{ so } a = \sqrt{3} \text{ m s}^{-2}$$

For the acceleration of the box after P has been applied:

$$\mathrm{R}(\rightarrow): \ P\cos 60^\circ + 6\cos 30^\circ = 3 \times 2\sqrt{3}$$

So $P = 6\sqrt{3}$ N

8.

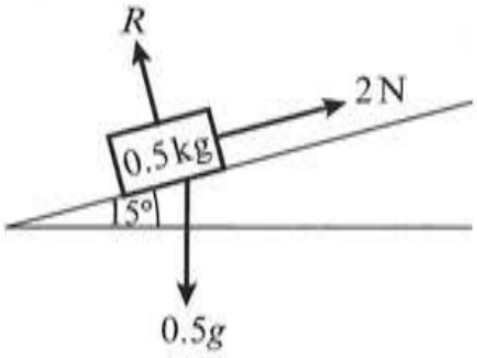
a. 41.4° (3 s.f.)

b. 6.75 (3 s.f.)

5.2 Inclined planes

1.
- **a.** $\frac{5\sqrt{3}}{2}g$ N
- **b.** $\frac{1}{2}g$ m s^{-2}

2. **a.**



- **b.** 1.46 m s^{-2} (3 s.f.)

3.
- **a.** 3.35 N (3 s.f.)
- **b.** 18.9 m s^{-1} (3 s.f.)

4. R(↙) : $mg\sin\theta = ma$ so $a = g\sin\theta$

5.
- **a.** 15.9 m s^{-1} (3 s.f.)
- **b.** 4.23 seconds (3 s.f.)

6. 6

7. $0.3g$ m s^{-1} down the slope.

8. 6.85 seconds

5.3 Friction

1. **a.**
- **i.** $0.4g$ N, 2.04 m s^{-2}
- **ii.** $0.7g$ N, 0.57 m s^{-2}

b. The frictional force will only be as large as it needs to be to oppose the 8 N force, so it will have a magnitude of 8 N.

2.
- **a.** 0.102 (3 s.f.)
- **b.** 0.412 (3 s.f.)
- **c.** 0.255 (3 s.f.)

3.
- **a.** 0.673 (3 s.f.)
- **b.** 2.06 (3 s.f.)
- **c.** 2.12 (3 s.f.)

4. 1.75 m s^{-2} (3 s.f.)

5.
- **a.** 80.4 m s^{-1} (3 s.f.)
- **b.** 20.5 seconds (3 s.f.)

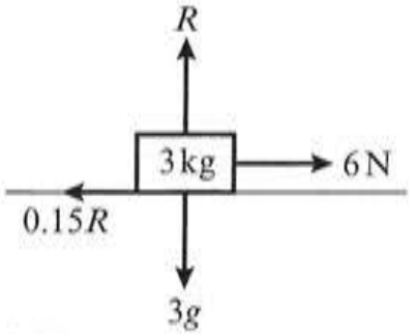
6. R(↖) : $R = mg\cos\theta$
R(↙) : $mg\sin\theta - \mu mg\cos\theta = ma$
So $a = g(\sin\theta - \mu\cos\theta)$

7.
- **a.** 2300 (3 s.f.)
- **b.** 580 (3 s.f.)

Problem solving: Set A

Bronze

a.



b. 9.54 m

Silver 11.3 m s^{-1}

Gold R(↑) : $R + 2g\sin\theta = mg$

So $R = mg - 2g\sin\theta$

R(→) : $2g\cos\theta - \mu(mg - 2g\sin\theta) = ma$

$$a = \frac{2g\cos\theta + 2g\mu\sin\theta}{m} - \mu g$$

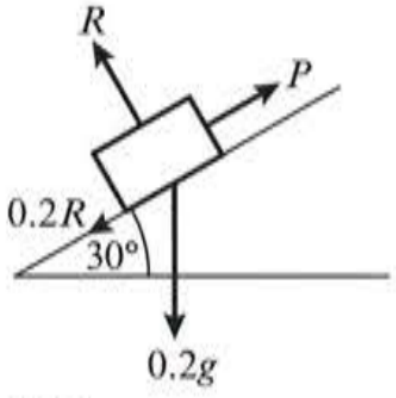
Using $v = u + at$ and $u = 0$ gives

$$v = \left(\frac{2\cos\theta + 2\mu\sin\theta}{m} - \mu\right) gt$$

Problem solving: Set B

Bronze

a.



b. $\frac{49\sqrt{3}}{250}$ N

c. 1.36 (3 s.f.)

Silver

a. R(↗) : $F > mg\sin\alpha + \mu R$
$F > \frac{3}{5}mg + 0.1 \times \frac{4}{5}mg$
$F > \frac{17}{25}mg$

b. 6.44 (3 s.f.)

Gold $\alpha = 13.7°$ or $30.7°$

Selected exam question bank answers

Question 22 6.91 kg

Question 25

a. 203 (3 s.f.)
b. P would be 0 as there would be no frictional force acting on the particle.

Question 29

a. 29.4
b. 0.482 (3 s.f.)

Question 32

a. 9.98 N (3 s.f.)
b. 22.6° (3 s.f.)

Question 36 261 N (3 s.f.)

Question 47

a. 0.877 (3 s.f.)
b. 2.68 m (3 s.f.)